

Flow-KTO: Learning Flow-Based Robot Policies from Unary Feedback

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Abstract

Flow-based policies are increasingly used to model multimodal, high-dimensional action distributions in robotic manipulation. Yet post-training them from real-world feedback faces two coupled obstacles: naive backpropagation through the multi-step generation process can yield unstable gradients, while, in the deployment setting we study, each physical observation yields only one executed rollout rather than same-context alternatives. We introduce Flow-KTO, a direct post-training objective that turns each desirable or undesirable trajectory label into a bounded, reference-relative update in conditional flow-matching space, centered on actions sampled from the current policy. Flow-KTO learns from independently labeled trajectories without fitting a reward or value model or collecting additional environment interactions.

1 Introduction

A robot deployment produces more than demonstrations. It also produces judgments about actions that should be repeated or avoided. In the setting we study, a policy observes a context, emits one action chunk, and changes the physical scene. The resulting trajectory can later be labeled desirable or undesirable, but the original context is gone: collecting a counterfactual action for exactly the same observation would require resetting objects, contact state, sensing, and timing to a condition that may not be reproducible. The natural training record is therefore a triple consisting of a logged context, one policy trajectory, and one unary quality label. Its unary structure is an operational constraint, not a synthetic removal of paired data.

That record does not instantiate the estimators used by several common post-training routes. Canonical Direct Preference Optimization (DPO) compares a preferred and a dispreferred response for the same context [1]. Proximal Policy Optimization (PPO) instead updates from rewarded policy trajectories and estimated advantages, often using a learned value function and generalized advantage estimation [2, 3]; it does not require preference pairs. Group Relative Policy Optimization (GRPO) replaces that value baseline with relative rewards from a group of sampled candidates [4]. Recent robot-learning systems also improve pretrained policies through learned rewards, values, or advantages, but their supervision includes richer temporal labels, autonomous rollouts, corrections, or transition-level outcomes [5, 6, 7]. Each route is viable when its information is available. A fixed set containing one binary-labeled trajectory per context provides neither a DPO pair, a rewarded PPO trajectory from the current policy, a rewarded GRPO group, nor the transition-level return targets and coverage assumed by a critic-based alternative.

The policy interface creates a separate obstacle. Flow and diffusion policies represent multimodal continuous actions through an iterative generative process [8, 9]. A sampled action is obtained by integrating a learned vector field from noise; its terminal log density is not a scalar emitted by the policy network. This does not make reinforcement learning impossible. Denoising-Markov-decision-process methods can optimize local stochastic transitions, and Flow Policy Optimization (FPO) constructs an ELBO-related ratio from conditional flow-matching (CFM) losses for on-policy PPO updates [10]. Those formulations have rewarded rollouts and advantages. Here the unresolved question is different: how can one update the terminal robot policy directly from a fixed unary label, using quantities native to flow training?

Two objects are required. First, the labeled trajectory needs a differentiable score that says how the current policy has changed relative to a fixed reference. Second, that score needs a reference point evaluated where the current policy places probability mass. The latter requirement is easy to miss. Scoring the logged action measures relative fit on the behavior data; scoring an action borrowed from another context measures

sensitivity to a context-action mismatch. Neither quantity asks what the current policy itself would produce at the logged observations. A policy-centered reference does. It can be estimated without touching the environment: sample an action from the model at a stored context, then teacher-force that same action through the current and reference flow models.

Flow-KTO follows this principle. For a labeled pair (x, y) , it defines a reference-relative improvement score $\Delta_\theta(x, y) = \mathcal{L}_{\text{CFM}}^{\text{ref}}(y | x) - \mathcal{L}_{\text{CFM}}^\theta(y | x)$. It then samples actions from the current policy at logged contexts and averages their corresponding scores to form a detached, dynamic surrogate reference point. A desirable label asks the logged score to exceed this center; an undesirable label asks it to fall below the center. Flow-KTO optimizes the resulting signed margin with the exact KTO utility, $1 - \sigma(\cdot)$, rather than a log-sigmoid or softplus classification loss [11]. The sampled actions receive no reward label and never enter the robot. They serve only to locate the current policy relative to its frozen reference.

This construction also exposes a numerical issue that is hidden by likelihood-based notation. CFM differences in our setting typically occupy the 10^{-3} to 10^{-2} range. At a 10^{-2} -scale inverse temperature, the KTO logit βm stays near zero and its utility changes almost imperceptibly with the margin. We therefore keep the KTO mixing and class-utility weights fixed while moving β to the 10^2 order, where the sigmoid responds to the observed score scale. This is not an extra loss weight. It is a calibration of the margin axis. For exact KTO, the resulting gradient is largest near the policy-centered boundary and attenuates on both saturated sides, retaining the bounded dynamic regulation of the original utility.

Several nearby methods establish important parts of the landscape. KTO introduced unpaired desirable/undesirable optimization and a theoretical policy-centered reference point for explicit language-model likelihood ratios [11]. Diffusion-KTO brought unary utility optimization to diffusion models through local reverse-transition ratios [12]; RKO has already evaluated unary alignment for real-world visuomotor diffusion policies [13]. FPO relates weighted CFM-loss differences to an ELBO ratio, but uses rewarded on-policy rollouts and advantages [10]. CrossVLA uses negative CFM error as a surrogate inside paired DPO [14], while MoRE edits trajectory modes using a temporary classifier and retain/redirect objectives [15]. These works preclude a novelty claim for unary utility or CFM differences in isolation. The unresolved optimization object is a flow-policy update for fixed operational logs that centers each unary CFM score on the current policy itself. Flow-KTO targets that object directly: one independently labeled trajectory per context, no learned reward/value critic, and no new environment interaction.

A controlled evaluation must isolate the objective, the reference estimator, and score-scale choices, and must test whether the updated policy preserves the competence of its pretrained reference. Those tests define the burden of proof. Our technical contributions are:

- We formulate direct post-training of a flow-based robot policy from a fixed set of unary-labeled trajectories, without paired alternatives, rewarded groups, a learned critic, or new environment interaction.
- We realize a policy-centered dynamic reference in the flow-training space by sampling current-policy actions at logged contexts and teacher-forced scoring those same actions under current and frozen reference policies. This estimator is formally distinct from scoring dataset actions or mismatched context-action pairs.
- We retain exact bounded KTO while matching its inverse temperature to the CFM-margin scale with fixed utility and objective weights. The resulting derivative makes the active margin band, peak update, and two-sided saturation explicit.

2 Method

2.1 Fixed Unary Robot Logs

Let $x \in \mathcal{X}$ denote the context available to a robot policy, including the observation history and any task conditioning, and let $y \in \mathbb{R}^{H \times d_a}$ be an action chunk of horizon H . The unary post-training set is

$$\mathcal{D}_u = \{(x_i, y_i, s_i)\}_{i=1}^N, \quad s_i \in \{+1, -1\}, \quad (1)$$

where $s_i = +1$ marks a desirable trajectory and $s_i = -1$ an undesirable one. There is at most one logged trajectory for a context in the studied corpus; no paired counterfactual y'_i is assumed.

We start from a pretrained flow policy π_{ref} and initialize the trainable policy π_{θ} from the same parameters. The reference remains frozen. Post-training does not query an environment, and internally sampled actions have no reward or quality label. We also allow an optional anchor set $\mathcal{D}_a$ of ordinary behavior-cloning trajectories. It is used only through the original CFM training objective and is kept separate from the unary feedback in Eq. (1).

2.2 A Reference-Relative CFM Improvement Score

We first define how a complete action chunk is scored by the flow policy. Let $\epsilon \sim p_0$ be a base-noise sample and let $t \sim p(t)$ denote a flow time. A conditional probability path can be written as

$$y_t = \alpha_t y + \sigma_t \epsilon, \quad u_t(y, \epsilon) = \dot{\alpha}_t y + \dot{\sigma}_t \epsilon, \quad (2)$$

where u_t is the conditional target velocity. This notation includes the linear optimal-transport path as a special case. For a vector field v_j with $j \in \{\theta, \text{ref}\}$, we use the same schedule, action mask M , and reduction as the pretrained policy:

$$\ell_j(t, \epsilon; x, y) = \frac{w(t)}{\|M\|_1} \|M \odot (v_j(y_t, t | x) - u_t(y, \epsilon))\|_2^2, \quad (3)$$

$$\mathcal{L}_{\text{CFM}}^j(y | x) = \mathbb{E}_{t, \epsilon} [\ell_j(t, \epsilon; x, y)]. \quad (4)$$

All current/reference comparisons share the same draws (t, ϵ) , masks, preprocessing, and normalization. Pairing removes nuisance variation from the difference without changing its expectation.

The Flow-KTO score is

$$\Delta_{\theta}(x, y) = \mathcal{L}_{\text{CFM}}^{\text{ref}}(y | x) - \mathcal{L}_{\text{CFM}}^{\theta}(y | x). \quad (5)$$

A positive value means that the current vector field fits the scored trajectory better than the frozen reference under the shared teacher-forced protocol. With R Monte Carlo draws, we estimate Eq. (5) by

$$\widehat{\Delta}_{\theta}^{(R)}(x, y) = \frac{1}{R} \sum_{r=1}^R [\ell_{\text{ref}}(t_r, \epsilon_r; x, y) - \ell_{\theta}(t_r, \epsilon_r; x, y)]. \quad (6)$$

The score has a precise but limited likelihood interpretation. For weightings under which the denoising objective satisfies $\mathcal{L}_j^w(y | x) = -\text{ELBO}_j(y | x) + C(x, y)$, as used in the FPO derivation [10],

$$\Delta_{\theta}^w(x, y) = \text{ELBO}_{\theta}(y | x) - \text{ELBO}_{\text{ref}}(y | x) \quad (7)$$

$$= \log \frac{\pi_{\theta}(y | x)}{\pi_{\text{ref}}(y | x)} - (G_{\theta}(x, y) - G_{\text{ref}}(x, y)), \quad (8)$$

where G_j is the model-dependent gap between log likelihood and its ELBO. Equation (8) is why a CFM difference can carry policy-improvement information, and also why we do not identify it with an exact log-likelihood ratio. Throughout, Δ_{θ} is a reference-relative CFM, or ELBO-related, improvement score.

2.3 A Policy-Centered Dynamic Surrogate Reference

Whether a labeled score is large or small depends on how the current policy has moved away from its reference. We therefore center it on current-policy actions. Let $q_{\mathcal{D}_u}(x)$ be the empirical context distribution. The target reference quantity is

$$b_{\theta} = \mathbb{E}_{x \sim q_{\mathcal{D}_u}} \mathbb{E}_{\tilde{y} \sim \pi_{\theta}(\cdot | x)} [\Delta_{\theta}(x, \tilde{y})]. \quad (9)$$

This expectation follows the actions that π_{θ} currently proposes. It does not assign utility to those actions.

For a batch of B logged contexts and K samples per context, the practical estimator is

$$\widehat{b}_{\mathcal{B}} = \text{sg} \left[\frac{1}{BK} \sum_{i=1}^B \sum_{k=1}^K \widehat{\Delta}_{\theta}^{(R)}(x_i, \tilde{y}_{ik}) \right], \quad \tilde{y}_{ik} \sim \pi_{\theta}(\cdot | x_i). \quad (10)$$

Sampling is performed without gradients. The sampled action is then teacher-forced under both policies using paired CFM randomness, and the entire aggregate is detached. In distributed training, workers aggregate the numerator and count before applying the stop-gradient. Larger K , larger R , or aggregation across batches can reduce variance; these choices affect estimator quality and compute but not the estimand in Eq. (9).

Two superficially similar quantities answer different questions. A dataset center

$$b_{\text{data}} = \mathbb{E}_{(x,y) \sim \mathcal{D}_u} [\Delta_{\theta}(x, y)] \quad (11)$$

measures relative fit on logged actions. A mismatched center such as $\mathbb{E}_i[\Delta_{\theta}(x_i, y_{\rho(i)})]$, for a nontrivial permutation ρ , measures response to context-action mismatch and may score out-of-distribution actions. Both are useful diagnostics. Neither is interchangeable with the current-policy expectation in Eq. (9). Because Δ_{θ} is a surrogate score, b_{θ} is likewise a dynamic surrogate reference point, not an exact KL divergence.

2.4 Exact KTO from a Signed Unary Margin

For a labeled example, define the signed policy-centered margin

$$m_i = s_i \left(\widehat{\Delta}_{\theta}^{(R)}(x_i, y_i) - \widehat{b}_{\mathcal{B}} \right). \quad (12)$$

Let λ_D and λ_U be the desirable and undesirable utility weights, and write $\lambda_{s_i} = \lambda_D$ for $s_i = +1$ and $\lambda_{s_i} = \lambda_U$ for $s_i = -1$. The per-example value and loss are

$$v_i(\theta) = \lambda_{s_i} \sigma(\beta m_i), \quad (13)$$

$$\ell_{\text{KTO},i}(\theta) = \lambda_{s_i} - v_i(\theta) = \lambda_{s_i} [1 - \sigma(\beta m_i)], \quad (14)$$

$$\mathcal{L}_{\text{KTO}}(\theta) = \mathbb{E}_{(x_i, y_i, s_i) \sim \mathcal{D}_u} [\ell_{\text{KTO},i}(\theta)]. \quad (15)$$

This is the exact bounded KTO utility [11]. There is no logarithm around the sigmoid. Written separately, the two cases are

$$\ell_D = \lambda_D [1 - \sigma(\beta(\Delta_{\theta} - b_{\theta}))], \quad (16)$$

$$\ell_U = \lambda_U [1 - \sigma(\beta(b_{\theta} - \Delta_{\theta}))]. \quad (17)$$

The sign convention gives the intended CFM update directly. Treating the reference point as stopped-gradient,

$$\frac{\partial \ell_{\text{KTO},i}}{\partial \widehat{\Delta}_i} = -s_i \lambda_{s_i} \beta \sigma(\beta m_i) [1 - \sigma(\beta m_i)]. \quad (18)$$

Minimization therefore increases Δ_i for a desirable example, which reduces its current-policy CFM loss relative to the reference. For an undesirable example it decreases Δ_i , increasing the current-policy CFM loss on that trajectory. The multiplier in Eq. (18) vanishes on both saturated sides; an example that is far from the boundary receives little update from this term.

2.5 Matching KTO to the CFM Margin Scale

Three coefficients play different roles. The class weights λ_D, λ_U set the bounded utility amplitudes and class balance. An outer coefficient η_{KTO} controls how the unary objective is mixed with an optional retention objective. The inverse temperature β sets the width of the active margin region. Conflating these controls would turn score calibration into a change in the total KTO weight.

For exact KTO, the magnitude of the margin derivative is

$$\left| \frac{\partial \ell_{\text{KTO},i}}{\partial m_i} \right| = \lambda_{s_i} \beta \sigma(\beta m_i) [1 - \sigma(\beta m_i)]. \quad (19)$$

Its peak is $\lambda_{s_i} \beta / 4$ at $m_i = 0$, and its active band has width on the order of $1/\beta$. In our flow-policy training, raw centered CFM margins are typically on the 10^{-3} to 10^{-2} scale. With $\beta = \mathcal{O}(10^{-2})$, a 10^{-3} margin gives $|\beta m| = \mathcal{O}(10^{-5})$; the utility remains effectively fixed at its midpoint. We instead keep λ_D, λ_U , and

η_{KTO} unchanged and use $\beta = \mathcal{O}(10^2)$. The sigmoid then resolves changes at the observed CFM scale, and its gradient automatically attenuates as either signed margin saturates.

Changing β does raise the peak boundary gradient even though the utility amplitude stays fixed. The optimizer learning rate, gradient clipping, score normalization, and sampling variance must therefore be controlled separately. The scale-matching claim is about the argument of the exact KTO utility; it does not imply that a larger β alone guarantees a better policy.

2.6 Training Objective and Procedure

The core objective is Eq. (15). When retention data are available, we optionally add the original flow-training loss

$$\mathcal{L}_{\text{anchor}}(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}_a} [\mathcal{L}_{\text{CFM}}^\theta(y | x)], \quad (20)$$

$$\mathcal{L}_{\text{total}}(\theta) = \eta_{\text{KTO}} \mathcal{L}_{\text{KTO}}(\theta) + \eta_a \mathcal{L}_{\text{anchor}}(\theta). \quad (21)$$

Setting $\eta_a = 0$ gives the pure unary objective. The anchor is a retention mechanism, not an additional source of preference or reward; its effect must be evaluated independently.

One optimization step proceeds as follows:

1. Draw a unary batch from $\mathcal{D}_u$ and, if used, an anchor batch from $\mathcal{D}_a$.
2. At each logged context, sample K action chunks from π_θ without gradient tracking and without querying the robot.
3. Teacher-force every sampled chunk through π_θ and π_{ref} with paired flow times and noise. Aggregate Eq. (10) across the batch and workers, then detach it.
4. Score the logged unary trajectories with Eq. (6), form their signed margins, and compute the exact KTO loss in Eq. (15).
5. Add the optional anchor term and update θ . The reference policy, policy samples, and dynamic reference receive no gradient.

For B contexts, K policy samples, S flow-solver steps, and R paired CFM draws, reference estimation adds BKS no-gradient vector-field evaluations for sampling and $2BKR$ teacher-forced current/reference evaluations, in addition to scoring the labeled examples. No reward model, value model, or backpropagation through the sampler is required. This cost is model computation, not environment interaction.

Flow-KTO assumes a fixed reference policy, binary trajectory-level judgments, and a consistent CFM scoring protocol whose numerical scale can be measured. It does not recover an exact terminal policy likelihood, and its dynamic reference is not an exact KL divergence. Nor does the formulation rule out critic-based offline learning or on-policy flow optimization when their richer signals are available. It addresses the narrower case established at the start: direct post-training from fixed unary robot logs.

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